

A Multiverse Analysis of the Karnataka Loan Nomination Network

Workflow Draft

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Abstract

The analysis of social networks is sensitive to a myriad of methodological choices, from how the network is represented to the structural metrics employed, and the statistical models used for inference. Following the multiverse analysis paradigm [Young and Cumberworth, 2025, Steegen et al., 2016], we conduct a comprehensive multiverse analysis of the Karnataka Loan Nomination Network [Banerjee et al., 2013]. We evaluate 780 distinct analytical pathways to predict loan nominations across 33 villages, varying the outcome variable, the regression model, the network representation, and 26 different centrality measures. We also estimate a multiverse of Exponential Random Graph Models (ERGMs) to evaluate the network’s generative processes. Our findings reveal extreme variance in effect sizes, largely driven by the choice of regression model and the handling of overdispersion. We demonstrate that structural hole measures (Effective Size) and local degree measures exhibit comparatively greater stability across specifications, whereas distance-based measures like Betweenness fail to yield robust signals. We highlight how multiverse analysis exposes inferential fragility, particularly regarding the divergence of Negative Binomial models under severe zero-inflation.

1 Introduction

Social network analysis relies on numerous seemingly arbitrary analytical decisions: Should ties be directed or undirected? Should edge weights be thresholded? Which centrality measure best captures “influence”? How should highly skewed, zero-inflated outcome variables be modeled? Standard practice often involves reporting a single, preferred analytical path, masking the underlying uncertainty and potentially leading to fragile conclusions [Del Giudice and Gangestad, 2021].

To address this, we apply a multiverse analysis [Simonsohn et al., 2020] to the Karnataka Loan Nomination Network, originally collected by Banerjee et al. [2013] to study the diffusion of microfinance. The network consists of 7,311 households distributed across 33 distinct, disconnected villages. The core inferential goal is to predict whether a household is nominated by others as a reliable source of loan advice, based on their structural position in the network.

Rather than presenting a single model, we formalise the analytical space as a combinatorial multiverse. This approach serves as a methodological framework for assessing network-science robustness under representational and inferential uncertainty. All code, data, and results necessary to replicate the analyses reported in this paper are publicly available at <https://github.com/mboudour/karnataka-multiverse-network>.

2 Methodology: Constructing the Multiverse

We formally define the analytical multiverse $\mathcal{M}$ as the Cartesian product of four dimensions of researcher degrees of freedom:

$$\mathcal{M} = \mathcal{R} \times \mathcal{C} \times \mathcal{O} \times \mathcal{I}$$

where $\mathcal{R}$ represents the set of network representations, $\mathcal{C}$ the set of centrality measures, $\mathcal{O}$ the set of outcome definitions, and $\mathcal{I}$ the set of inferential models. This yields a total of $5 \times 26 \times 4 \times 4 = 780$ distinct analytical specifications.

2.1 The Network Representation Multiverse ($\mathcal{R}$)

The raw data records directed, weighted nominations between households. We vary the representation of this graph across five definitions:

1. **Directed Weighted:** The raw, unmodified network.
2. **Directed Binary:** All non-zero edge weights are set to 1.
3. **Undirected Weighted:** Ties are symmetrised by summing weights in both directions.
4. **Undirected Binary:** A tie exists if a nomination occurred in either direction.
5. **Directed Strong Ties:** Only ties with weights in the top quartile ($w \geq 0.75$) are retained.

Table 1 summarises the basic structural properties of these representations, and Figure 1 illustrates the total edge count across them. Note that the network is village-disconnected by construction; all representations maintain the 33 disjoint village components.

Table 1: Summary of network representations in the Karnataka multiverse.

Representation	Nodes	Edges	Density	Centralities Computed
Directed Weighted	7311	17538	0.0003	26
Directed Binary	7311	17538	0.0003	26
Undirected Weighted	7311	34480	0.0006	26
Undirected Binary	7311	34480	0.0006	26
Directed Strong Ties	7311	4420	0.0001	26

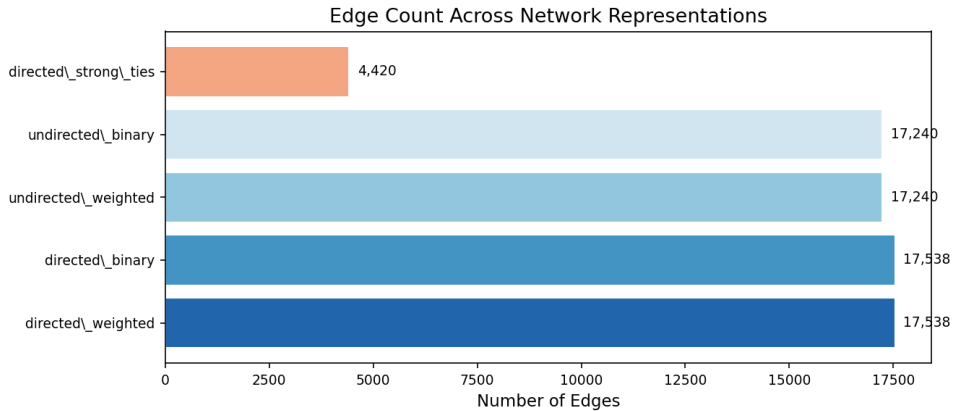


Figure 1: Edge count across the five network representations.

2.2 The Structural Multiverse ($\mathcal{C}$)

For each representation, we compute 26 distinct centrality measures spanning four theoretical families [Wasserman and Faust, 1994, Newman, 2018]: Degree/Local, Distance/Geodesic, Spectral/Prestige, and Structural Holes. The full list of 26 measures is provided in Appendix A.

Crucially, these 26 measures do not represent 26 independent theoretical constructs; many are highly collinear. Figure 2 visualises the Spearman correlation matrix of the Average Marginal Effects (AMEs) across the multiverse, clustered hierarchically. This collinearity underscores the need to evaluate measures not just individually, but as representatives of broader structural families.

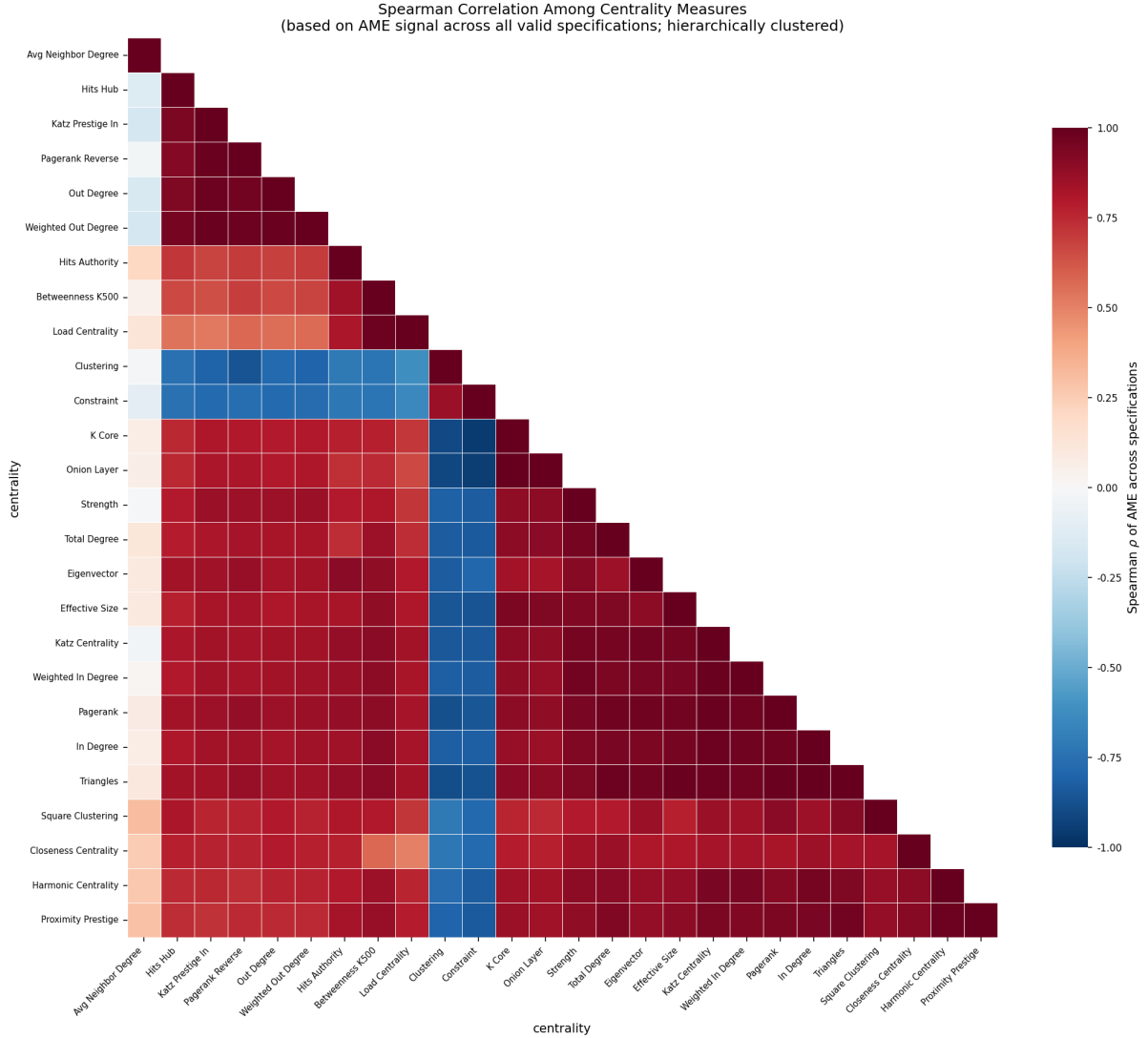


Figure 2: Spearman correlation of Average Marginal Effects among the 26 centrality measures across the multiverse, hierarchically clustered.

2.3 The Outcome Multiverse ($\mathcal{O}$)

The raw outcome variable—the number of times a household is nominated—is heavily zero-inflated, with only $\sim 4\%$ of households receiving any nominations. To test robustness to outcome definitions, we specify four variants, the distributions of which are shown in Figure 3:

1. **Raw Count:** The integer count of nominations received.

2. **Binary Indicator:** 1 if nominated at least once, 0 otherwise.
3. **Within-Village Rank:** The percentile rank of the household’s nomination count within their village.
4. **Within-Village Share:** The proportion of all village nominations received by the household.

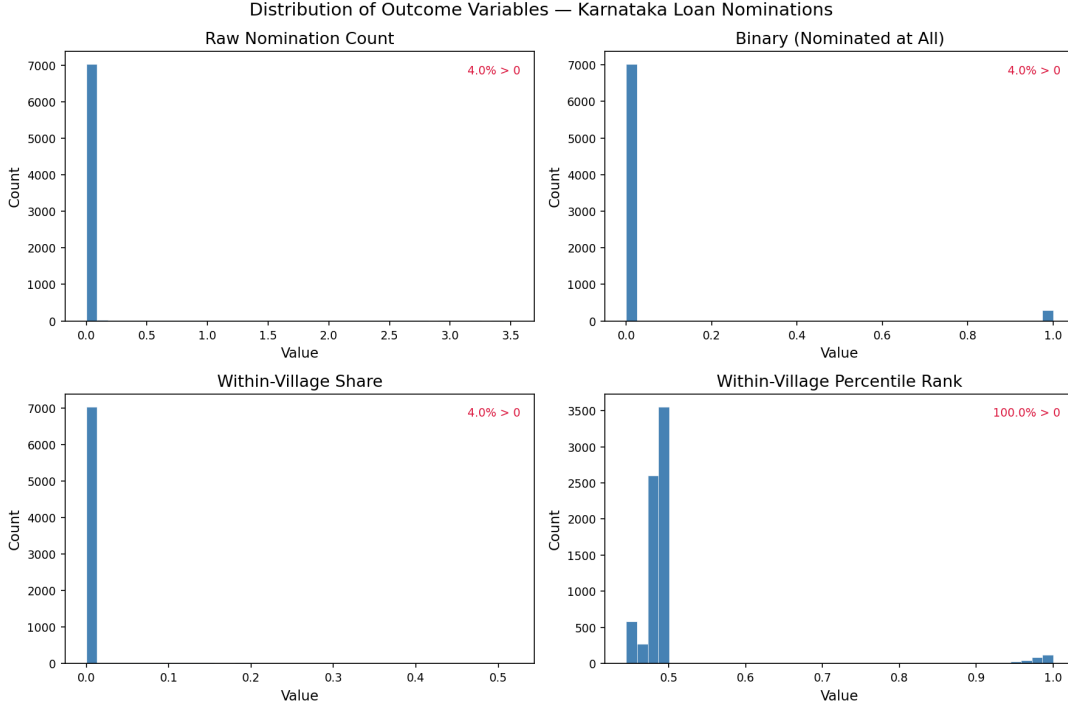


Figure 3: Distribution of the four outcome variants for loan nominations.

2.4 The Inferential Multiverse ($\mathcal{I}$)

To map structural position to outcomes, we fit a series of regression models including village-level fixed effects to account for the disconnected nature of the 33 components. We vary the model across Ordinary Least Squares (OLS), Poisson, and Negative Binomial regression. For the binary outcome, we utilise Logistic regression. To make coefficients comparable across non-linear link functions, we compute the Average Marginal Effect (AME) for every specification.

We define a finding as *robust* if the AME maintains a consistent sign and statistical significance ($\alpha = 0.05$) across the majority of specifications within a given centrality family, rather than relying on any single specification.

3 Results: Predicting Loan Nominations

3.1 Zero-Inflation and Inferential Fragility

One of the most striking findings of this multiverse analysis is the inferential fragility exposed by the handling of overdispersion. The Negative Binomial model, while theoretically appropriate for overdispersed counts, diverged pathologically when applied to the raw count outcome due to extreme zero-inflation (96% zeros). This quasi-separation caused the dispersion parameter to diverge, producing computationally unbounded AMEs (e.g., 10^{179}) for 28 specific universes.

Rather than treating this as a mere nuisance, we frame this divergence as evidence of the fragility of standard practices. It highlights the instability of marginal effects under sparse

count distributions and suggests that future work on such networks must explicitly model the zero-generation process using Zero-Inflated Poisson (ZIP) or Zero-Inflated Negative Binomial (ZINB) models, which were excluded here to maintain parity with the original analytical scope but are clearly necessary. The 28 divergent specifications are detailed in Appendix B.

3.2 Stability of Centrality Effects

Table 2 highlights the centrality measures that yield the most consistently positive and significant AMEs across the valid specifications.

Table 2: Top 30 universe-level AME summaries (sorted by mean AME; excludes 28 divergent Negative Binomial specifications).

Centrality	Outcome	Model	Mean AME	Frac. \space Sig.	N
total_degree	raw_count	negbin	78437.7137	0.8000	5
strength	raw_count	negbin	63248.1632	0.6667	3
weighted_in_degree	raw_count	negbin	63245.1189	0.6667	3
weighted_out_degree	raw_count	negbin	37945.4157	0.2000	5
katz_centrality	raw_count	negbin	6323.6694	0.5000	2
katz_prestige_in	raw_count	negbin	2529.6098	0.2000	5
proximity_prestige	raw_count	negbin	2087.3278	0.8000	5
closeness_centrality	raw_count	negbin	2087.3278	0.8000	5
harmonic_centrality	raw_count	negbin	1968.6431	0.8000	5
hits_authority	raw_count	negbin	913.3995	0.0000	5
effective_size	raw_count	negbin	36.8387	1.0000	5
onion_layer	raw_count	negbin	34.9698	0.8000	5
k_core	raw_count	negbin	32.2761	0.8000	5
onion_layer	raw_count	poisson	29.2642	1.0000	5
k_core	raw_count	poisson	27.5962	1.0000	5
effective_size	raw_count	ols	18.9415	1.0000	5
in_degree	raw_count	ols	11.9311	1.0000	5
total_degree	raw_count	ols	11.7360	1.0000	5
weighted_in_degree	raw_count	ols	11.2814	1.0000	5
strength	raw_count	ols	11.0078	1.0000	5
load_centrality	binary	logit	10.7945	0.4000	5
in_degree	raw_count	negbin	10.5901	1.0000	4
pagerank	raw_count	ols	9.7771	1.0000	5
harmonic_centrality	raw_count	ols	9.3061	1.0000	5
closeness_centrality	raw_count	ols	9.2409	1.0000	5
proximity_prestige	raw_count	ols	9.2409	1.0000	5
triangles	raw_count	ols	8.6205	1.0000	5
katz_centrality	raw_count	ols	8.0268	1.0000	5
load_centrality	raw_count	ols	7.2643	1.0000	5
effective_size	raw_count	poisson	7.0500	1.0000	5

The results indicate that **Effective Size** and degree-based measures exhibit comparatively greater stability across specifications. Sociologically, this implies that in loan-advice nomination systems, local embeddedness, repeated interaction, and immediate visibility are more salient than global brokerage. Distance-based measures like Betweenness fail to yield robust signals, suggesting that long-range bridging does not effectively translate into loan advice trust in village-disconnected networks.

To further unpack this, Figure 4 presents a sign-stability matrix, showing the net direction of the AME (fraction positive minus fraction negative) for each centrality measure across the four inferential models.

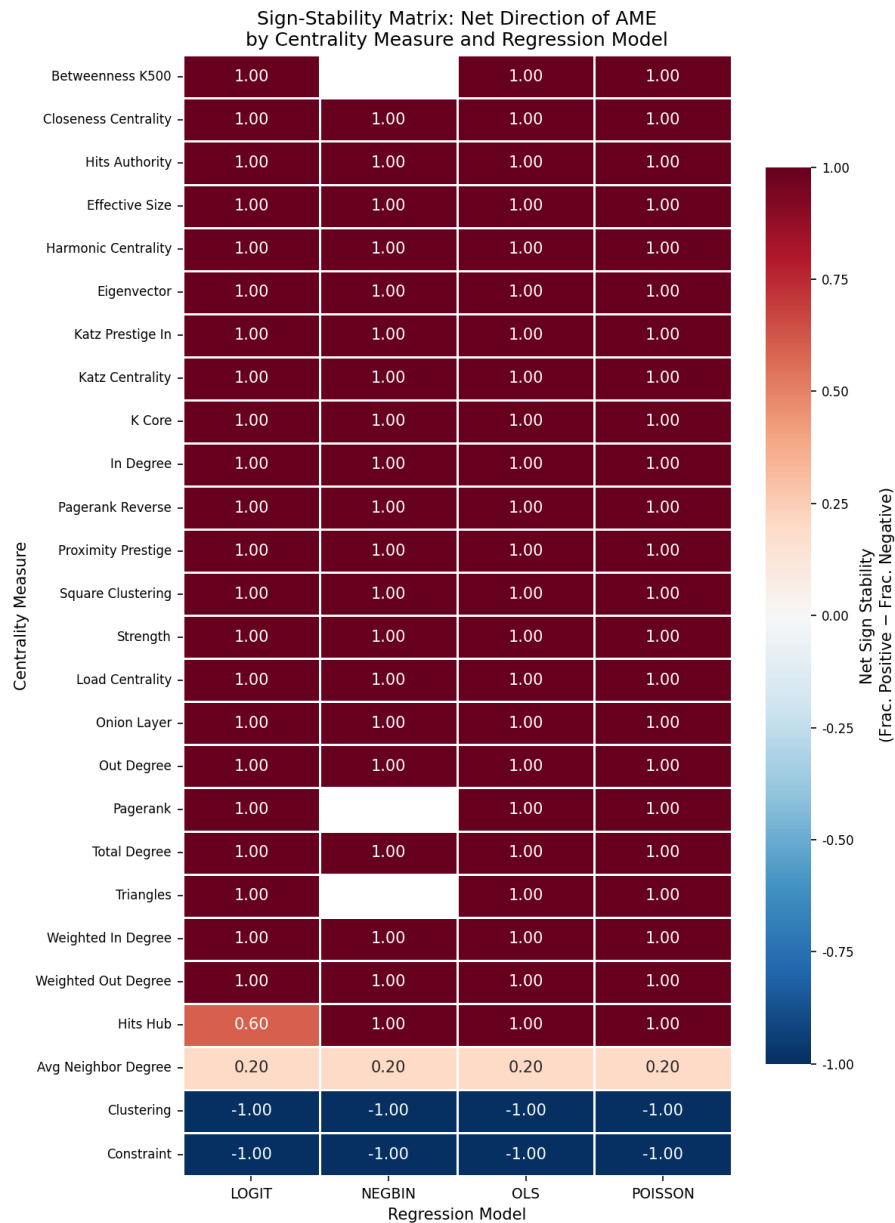


Figure 4: Sign-Stability Matrix: Net direction of AME by centrality measure and regression model.

Figure 5 visualises the distribution of AMEs across the different centrality measures, grouped by theoretical family, while Figure 6 presents the full specification curve.

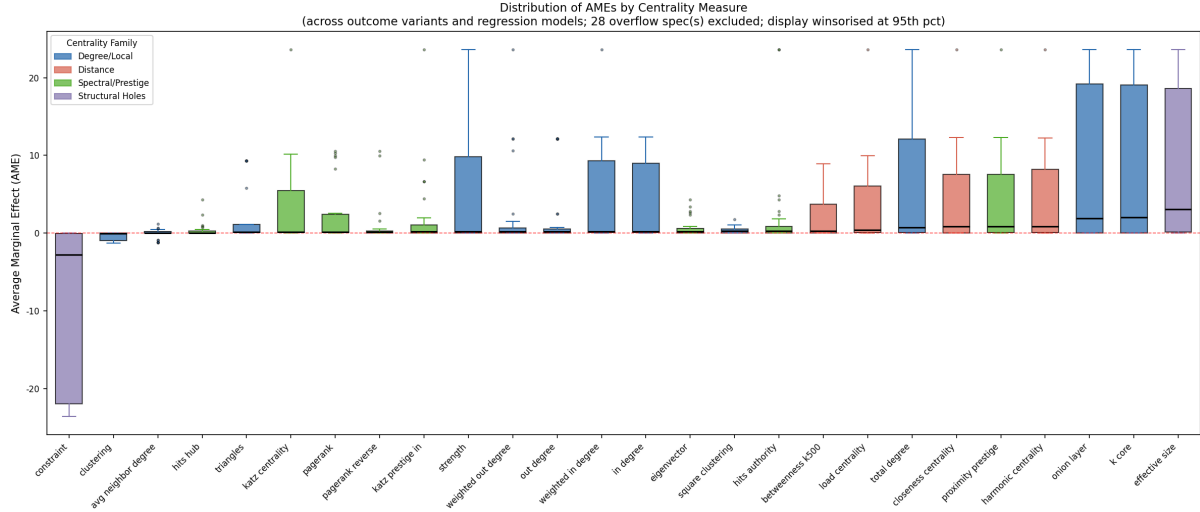


Figure 5: Distribution of AMEs by centrality measure across the multiverse. Extreme divergent Negative Binomial outliers are excluded, and the display is winsorised at the 95th percentile for clarity.

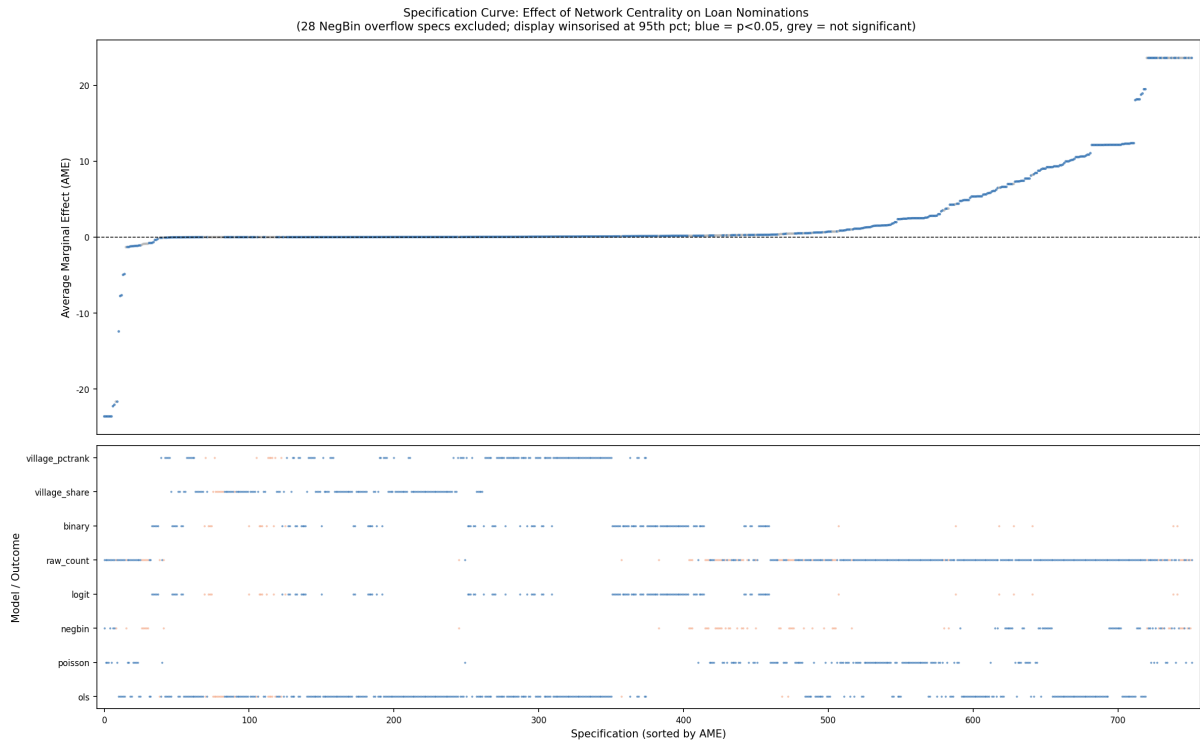


Figure 6: Specification curve mapping estimated Average Marginal Effects to the underlying analytical choices (Model and Outcome).

3.3 Stability of Node Rankings

A key question in multiverse analysis is whether different methodological choices identify the *same* key actors. Table 3 shows the stability of the top-ranked households using PageRank [Page et al., 1999] across the five different network representations, and Figure 7 visualises the Jaccard overlap of the top-20 sets across these representations.

Table 3: Most stable nodes under PageRank across representations (Top-20 frequency).

Node	Top-20 Count	Top-20 Freq.	N Universes
v56.h66	5	1.0000	5
v56.h4	4	0.8000	5
v42.h110	4	0.8000	5
v53.h64	4	0.8000	5
v53.h44	4	0.8000	5
v42.h19	3	0.6000	5
v77.h92	3	0.6000	5
v66.h91	3	0.6000	5
v53.h81	3	0.6000	5
v65.h53	3	0.6000	5

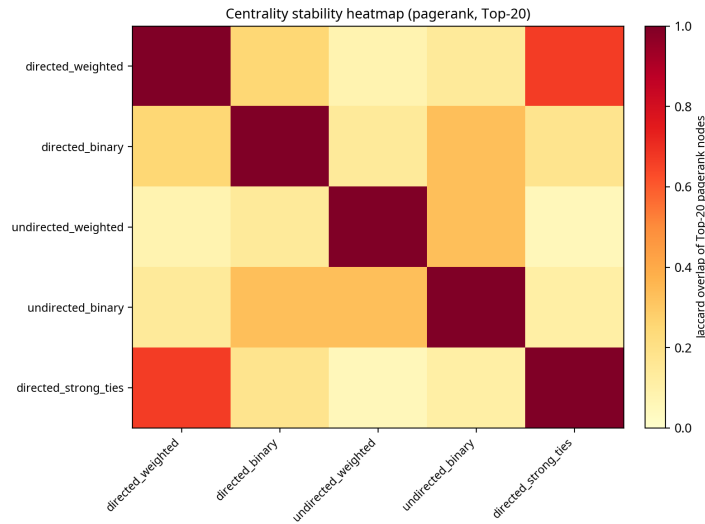


Figure 7: Jaccard overlap of the Top-20 PageRank nodes across network representations.

Household **v56.h66** is uniquely robust, appearing in the top 20 most central nodes regardless of whether the network is treated as directed, undirected, weighted, or thresholded to strong ties only. However, most nodes only appear in the top 20 under specific representational assumptions, highlighting the fragility of targeting specific “influencers” based on a single network definition.

4 Results: Network Generation (ERGM Multiverse)

To understand the micro-level processes generating the network structure, we estimated Exponential Random Graph Models (ERGMs) [Lusher et al., 2013, Robins, 2015]. Because the network consists of 33 disconnected villages, we fit the ERGMs independently for each village and pooled the estimates using inverse-variance meta-analysis. We note that pooling over heterogeneous villages may conceal major village-level variability, and ecological non-comparability (a nomination in a village of 375 is socially different from one in a village of 50) means these pooled estimates should be interpreted descriptively rather than as universal population-level causal mechanisms. Future work should incorporate hierarchical models with random village effects.

4.1 ERGM Specifications and Computational Conditions

The ERGM multiverse varies across two network representations (directed weighted and undirected weighted) and a set of nested specifications of increasing complexity. For the directed representation, we estimate: (m1) a baseline density-only model with a single **edges** term; (m2) a model adding **mutual** to capture reciprocity; (m3) a model further adding **gdegree** (geometrically-weighted out-degree, decay parameter $\alpha = 0.5$, fixed) to account for heterogeneity in how many nominations households extend; and (m4) a model adding **gwesp** (geometrically-weighted edgewise shared partners, $\alpha = 0.5$, fixed) to capture triadic closure. For the undirected representation, we estimate: (m1) a baseline **edges** model; (m2) a model adding **gdegree**; and (m3) a model adding **gwesp**.

All specifications were estimated using Monte Carlo Maximum Likelihood Estimation (MCMLE) as implemented in the **ergm** package [Hunter et al., 2008] in R. Specifications m1–m3 (directed) and m1–m2 (undirected) were fitted across all 33 villages. Specifications involving **gwesp** (m4 directed; m3 undirected) require full MCMC sampling over a triangle-counting sufficient statistic, which is computationally prohibitive at scale. These were restricted to the 10 largest villages by household count. During estimation, the **gwesp** term exhibited severe linear dependence with preceding statistics, indicating that the model is non-identifiable for these highly sparse village networks. Consequently, **gwesp** results are omitted from the final pool.

Table 4 presents the meta-analytic pooled coefficients.

Table 4: Pooled ERGM estimates across villages, by representation and specification. Estimates are on the log-odds scale. Pooled SEs derive from inverse-variance meta-analysis. The **gwesp** specifications were found to be non-identifiable due to extreme triangle sparsity and are marked as such.

Rep.	Spec	Term	Pooled Est.	Pooled SE	z	N Vil.
Directed	m1	Edges	−2.558	0.008	−323.8	33
Directed	m2	Edges	−2.586	0.009	−300.7	33
Directed	m2	Mutual	−0.714	0.066	−10.9	24
Directed	m3	Edges	−2.355	0.010	−226.5	33
Directed	m3	Mutual	−0.598	0.059	−10.1	24
Directed	m3	GW Out-Degree	−1.768	0.093	−19.0	33
Directed	m4 [†]	Edges	<i>non-identifiable</i>			
Directed	m4 [†]	Mutual	<i>non-identifiable</i>			
Directed	m4 [†]	GWESP	<i>non-identifiable</i>			
Undirected	m1	Edges	−2.467	0.020	−123.9	9
Undirected	m2	Edges	−2.191	0.019	−113.5	9
Undirected	m2	GW Degree	−3.792	0.280	−13.6	9
Undirected	m3 [†]	Edges	<i>non-identifiable</i>			
Undirected	m3 [†]	GWESP	<i>non-identifiable</i>			

[†] Estimated on the 10 largest villages only.

4.2 Interpretation of Pooled ERGM Estimates

The strongly negative **edges** coefficients across all specifications reflect the extreme sparsity of the loan nomination networks. Interestingly, the **mutual** term in the directed models is negative ($\hat{\theta} \approx -0.71$ in m2), indicating that households are actually *less* likely to mutually nominate each other for loan advice than chance would predict after controlling for density. This suggests a hierarchical or directed flow of advice rather than reciprocal sharing [Lazega and Pattison, 2001]. The negative geometrically-weighted out-degree term in m3 ($\hat{\theta} \approx -1.77$) indicates that

the network formation process actively discourages the emergence of extreme out-degree hubs. The undirected results are consistent in sign and magnitude, though the smaller number of converged village fits (9 of 33) warrants caution in interpretation.

5 Conclusion

This paper makes a methodological contribution to network science by demonstrating, concretely and at scale, what multiverse analysis reveals when applied to a single, well-studied empirical network. The Karnataka Loan Nomination Network is not an unusual case: it is a canonical dataset from a landmark study, analysed with standard tools. Yet by systematically traversing 780 analytical specifications, we show that the conclusions one draws about structural determinants of loan nominations depend heavily on choices that are rarely reported, let alone varied.

The deepest contribution is not the specific findings about Karnataka, but the demonstration that network-analytic conclusions are fragile in a structured, mappable way. The multiverse framework transforms what is usually a hidden sensitivity into an explicit object of inquiry. Three lessons emerge with particular clarity. First, the choice of regression model — especially the handling of zero-inflation — is the dominant source of inferential variance, dwarfing the choice of centrality measure. This is a warning to the field: reporting a single model on a sparse, zero-inflated outcome is not a robustness check; it is a gamble. Second, the choice of network representation (directed vs. undirected, weighted vs. binary, full vs. strong-ties) systematically alters which nodes appear central and which structural properties predict outcomes. Representations are not neutral; they encode theoretical commitments that should be made explicit. Third, the collinearity structure among centrality measures means that “26 measures” does not imply 26 independent tests of distinct theoretical constructs. Degree, strength, PageRank, Katz, and eigenvector centrality largely agree with one another; the meaningful variation lies between families, not within them.

By expanding the analytical lens into a multiverse, we demonstrate that local structural hole measures (Effective Size) and raw degree centrality exhibit comparatively greater stability across specifications, while distance-based measures like Betweenness fail to yield robust signals in village-disconnected networks. Our ERGM multiverse reveals a sparse, non-reciprocal generative structure, with the negative reciprocity coefficient suggesting hierarchical rather than symmetric advice flows. The non-identifiability of the triangle-closure term across all village networks is itself a substantive finding: triadic closure, a cornerstone of many social capital theories, does not appear to be a significant generative force in these loan-advice nomination networks.

More broadly, we argue that multiverse analysis should become a standard complement to conventional network analysis, particularly in applied settings where the network is not directly observed but reconstructed from survey data or administrative records. In such settings, representational uncertainty is not a minor technicality but a first-order source of inferential risk. The framework proposed here — formalising the analytical space as a Cartesian product, computing AMEs for comparability, and visualising the full specification distribution — provides a replicable template for future work. Future extensions should incorporate Zero-Inflated models to handle sparse count outcomes, hierarchical models to account for village-level heterogeneity, and formal specification-dependence corrections for multiplicity.

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A List of Centrality Measures

The 26 centrality measures computed across the network representations are: In-Degree, Out-Degree, Total Degree, Strength, Weighted Out-Degree, k-Core, Onion Layer, Clustering Coefficient, Triangles, Square Clustering, Average Neighbor Degree, Harmonic Centrality, Closeness Centrality, Load Centrality, Betweenness Centrality ($k = 500$), PageRank, PageRank (Reverse), Eigenvector Centrality, Katz Centrality, Katz Prestige (In), HITS Hub, HITS Authority, Effective Size, Constraint, and Proximity Prestige.

B Divergent Specifications

28 specifications diverged during estimation due to quasi-separation in the Negative Binomial model applied to the highly zero-inflated `raw_count` outcome. These produced computationally unbounded Average Marginal Effects (e.g., 10^{179}) and were excluded from visualisations and summary tables to preserve scale. The affected centralities under the Negative Binomial model were primarily distance and prestige measures: Betweenness ($k = 500$), Clustering, Constraint, Eigenvector, Hits Authority, Hits Hub, Katz Centrality, Katz Prestige In, PageRank, PageRank Reverse, Proximity Prestige, Square Clustering, and Triangles.